Activity

1. Find the mean, median and mode of : 5, 7, 7, 9, 11

(i) Mean: $\frac{5+7+7+9+11}{5} = \frac{39}{5} = 7.8$

(ii) Median : 7

(iii) Mode : 7

2. Find the mean, median, mode and mid-range of : 12, 15, 15, 18, 24

(i) Mean = $\frac{12+15+15+18+24}{5} = \frac{84}{5} = 16.8$

(ii) Median = 15

(iii) Mode = 15

(iv) Mid range = $\frac{24+12}{2} = 18$

3. Find mean, variance and standard deviation of 4, 6, 8, 10, 12

(i) Mean = $\frac{4+6+8+10+12}{5} = 6.4$

(ii) Variance =

value (x)	x - Mean	$(x - \text{mean})^2$	
4	-2.4	5.76	
6	-0.4	0.16	
8	1.6	2.56	Variance = 10.56
10	3.6	12.96	
12	5.6	31.36	

(iii) Standard deviation = $\sqrt{10.56} = 3.2496$

4) Find the mean, median, variance and standard deviation of 8, 10, 12, 14, 16

(i) Mean = $\frac{8+10+12+14+16}{5} = 12$

(ii) Median : 12

(iii) variance :

x value	x - mean	(x - mean)^2
8	-4	16
10	-2	4
12	0	0
14	2	4
16	4	16

variance = 8

(iv) Standard deviation : $\sqrt{8} = 2.828$

5). Find the mean, median, mode, variance, Standard deviation of 3, 5, 5, 7, 10

(i) Mean = $\frac{3+5+5+7+10}{5} = 6$

(ii) Median = 5

(iii) Mode = 5

(iv) variance =

x value	(x - mean)	(x - mean)
3	-3	9
5	-1	1
5	-1	1
7	1	1
10	4	16

variance = 5.6

(v) Standard deviation : $\sqrt{5.6} = 2.366$

6). Find the mean, median, and mid range of 18, 20, 22, 24, 26

(i) Mean = $18+20+22+24+26 / 5 = 22$

(ii) Median = 22

(iii) Mid-range = $\frac{26+18}{2}$ = 22

7) Find the mean, mode, variance and standard deviation of: 6, 8, 8, 10, 12

(i) Mean = $\frac{6+8+8+10+12}{5}$ = 92

(ii) Mode = 8

(iii) variance =

x value	(x - mean)	(x - mean)^2
6	-16	256
8	-14	196
8	-14	196
10	-11	121
12	-10	100

variance = $\frac{256+196+196+121+100}{5} = \frac{869}{5}$ = 173.8

(iv) Standard deviation: $\sqrt{173.8}$ = 13.1833

8) Find mean, median and variance of 15, 18, 21, 24, 27

(i) Mean = $\frac{15+18+21+24+27}{5} = \frac{105}{5}$ = 21

(ii) Median = 21

(iii) variance =

x value	(x - mean)	(x - mean)^2
15	-6	36
18	-3	9
21	0	0
24	3	9
27	6	36

variance = $\frac{36+9+0+9+36}{5} = \frac{90}{5}$ = 18

9. Find the mean, median, mode and standard deviation of 9, 11, 11, 13, 15

(i) $\text{Mean} = \frac{9+11+11+13+15}{5} = \frac{59}{5} = 11.8$

(ii) Median = 11

(iii) Mode = 11

(iv) Standard Deviation:

x value	(x − mean)	(x − mean)^2
9	−2.8	7.84
11	−0.8	0.64
11	−0.8	0.64
13	1.2	1.44
15	3.2	10.24

$\text{variance} = \frac{20.8}{5} = 4.16$

Standard deviation $= \sqrt{4.16} = 2.039$

10) Find the mean, median, mode, mid-range, variance and standard deviation of 10, 12, 14, 14, 200

(i) $\text{Mean} = \frac{10+12+14+14+200}{5} = 50$

(ii) Median = 14

(iii) Mode = 14

(v) Mid range:

$\frac{200+10}{2} = \frac{210}{2} = 105$

(V) variance

x value	(x - mean)	(x - mean)^2
10	-40	1600
12	-38	1444
14	-36	1296
14	-36	1296
200	150	22500

$$\text{variance} = \frac{28136}{5} = 5627.2$$

(VI) Standard deviation: $\sqrt{5627.2} = 75.014$